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A Project Proposal On
Pharmalink A B2B ecommerce and supply chain management System

Submitted To:
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1. Introduction

In Nepal, the pharmaceutical supply chain between manufacturers, distributors, wholesalers, and pharmacies is still mostly handled through manual processes or disconnected systems. Many businesses rely on phone calls, emails, or personal visits to place orders and manage stock. This makes the process slow, inefficient, and prone to errors.

Because even when digital systems like e-LMIS [1] and private sector ai solutions exist they operate in isolation, this reliance on disconnected methods results in difficulties in tracking inventory, comparing suppliers, ensuring timely delivery of medicines, and processing payments securely. Small pharmacies and distributors also struggle to find reliable suppliers and manage bulk orders efficiently.

To solve this fragmentation problem, a new web-based system called PharmaLink is proposed. PharmaLink is a B2B e-commerce and supply chain management platform. it aims to unify key actors on a single digital system where suppliers can list medicines and buyers can place bulk orders digitally. The system allows users to view product details, check availability, compare prices, track orders in real time, and complete payments through integrated digital payment gateways.

2. Problem Statement

Despite some digitalization efforts, Nepal's pharmaceutical B2B sector continues to face several critical challenges:

1. **Fragmented Digital Systems:** Multiple systems exist (e-LMIS, private AI solutions), but no centralized B2B platform connects manufacturers, distributors, wholesalers, and pharmacies, leading to inefficiency.
2. **Manual and Slow Order Processing:** Many businesses still rely on phone calls or physical visits, causing delays and errors, especially outside major digital networks.
3. **Poor Inventory Visibility:** Lack of real-time stock level and expiry date tracking across the supply chain persists for many stakeholders.
4. **Lack of Smart Product Search:** Buyers struggle to quickly filter and compare medicines by brand, category, or price on available platforms.

5. **No Automated B2B Order Tracking:** Once an order is placed, digital tracking is often unavailable.
6. **Data and Security Issues:** Reliance on paper records or unintegrated digital systems creates data inconsistency and security risks.
7. **Limited Access for Small Businesses:** Small pharmacies face difficulties connecting efficiently with large suppliers and integrating into digital systems used by bigger players.
8. **No Integrated Digital Payment System:** Transactions often rely on cash-on-delivery or manual bank transfers, causing payment delays and reconciliation problems.

These limitations highlight the urgent need for a secure and centralized digital B2B pharmaceutical system like **PharmaLink**.

3. Objectives

The main objectives of the proposed PharmaLink system are as follows:

1. To design and develop a centralized B2B web-based platform that digitalizes the pharmaceutical supply chain in Nepal by enabling verified suppliers, distributors, and buyers to manage products, process bulk orders, and conduct secure transactions through integrated digital payment gateways.
2. To implement a structured business verification and inventory management system that ensures regulatory compliance, maintains real-time stock accuracy, and provides transparent order tracking across all stakeholders in the pharmaceutical trade network.

4. Methodology

PharmaLink will be developed using the **Agile software development methodology**, where the project is divided into small development phases (sprints). Each sprint will include planning, development, testing, and review. This approach allows continuous improvement of the system based on feedback and testing, helps in identifying errors early, and delivers a stable product gradually.

4.1. Sprint Plan

1. Sprint 1

- Requirement Analysis
- System Architecture & Design

2. Sprint 2

- Authentication and Database Setup
- Basic UI Development
- Testing

3. Sprint 3

- Product Management & Inventory System
- Order Placing and Tracking System
- Testing

4. Sprint 4

- Payment Gateway Integration
- Search Filtering and Dashboard
- Testing

5. Sprint 5

- Deployment & Final Testing

Documentation and reporting will be a continuous process throughout all sprints.

4.2. Requirement Identification

4.2.1. Study of Existing System / Literature Review

A comprehensive analysis of existing pharmaceutical supply chain systems in Nepal was conducted to identify operational gaps and innovation opportunities for PharmaLink.

4.2.1.1 Government Systems

- **e-LMIS (Electronic Logistic Management Information System):** Introduced in 2018 with USAID support, tracks medicine stocks in government health facilities with real-time visibility. Limited to public sector only.
- **SPARS Strategy (Supervisory, Performance Assessment and Recognition Strategy):** Implemented under USAID MTaPS program [2], improved medicine management in 12 districts but involves high operational costs and physical supervision.

4.2.1.2 Private Sector Solutions

- **Pharma X Nepal:** AI-driven inventory tracking, proprietary system limited to its own distribution network [3].
- **Pharma Life Network:** Connects over 2,000 pharmacies nationwide with AI-based logistics optimization, but restricted to affiliated pharmacies [4].
- **General B2B Platforms:** Platforms such as SastoDeal [5] and Hamrobazar [6] facilitate general trading but lack pharmaceutical-specific features.

4.2.1.3 Key Limitations Identified

1. System Fragmentation
2. Absence of Dedicated B2B Pharma Platform
3. Limited Payment Integration
4. Poor Real-Time Visibility
5. High Technical Barriers
6. Exclusion of Small Businesses

4.2.1.4 How PharmaLink Addresses These Gaps PharmaLink is designed according to Department of Drug Administration (DDA) standards [7] and incorporates modern web technologies with secure authentication and payment integration [8, 9]. It builds upon existing strengths while offering a unified, open-access B2B pharmaceutical trading platform.

4.3. Requirement Analysis

4.3.1. Functional Requirements

The functional requirements define the core operations and features that the PharmaLink system must support to enable efficient B2B pharmaceutical trading.

1. User Registration and Login (Admin, Supplier, Buyer)

- Suppliers (pharmaceutical companies and distributors), Buyers (hospitals, pharmacies, and clinics), and Administrators must be able to create user accounts through a verified registration process.
- All users shall be authenticated based on certified license verification to ensure regulatory compliance.
- The system must allow secure login using email or mobile number and password with proper session management.

2. Product Listing and Management

- Suppliers shall be able to add, update, and delete their products through a dedicated product management interface.
- Each product record must include essential pharmaceutical details such as:
 - Product name
 - Batch number
 - Expiry date
 - Available quantity
 - Unit price
 - Manufacturer details
- The system must ensure that expired products cannot be listed for sale.

3. Order Placement and Order Tracking

- Buyers shall be able to browse products and place orders based on their requirements.
- Once an order is placed, the system must generate a unique order ID and allow buyers to track the order status through various stages such as:
 - Order Confirmed
 - Processing
 - Shipped
 - Delivered
- Both buyers and suppliers shall receive real-time notifications regarding order updates.

4. Inventory Management System

- The system must automatically update stock levels after each order is placed or fulfilled.
- Suppliers shall be notified when stock levels fall below a predefined threshold.
- The system shall prevent orders from being placed for out-of-stock products and maintain real-time inventory visibility.

5. Smart Search and Product Filtering

- The platform must provide an advanced search feature that allows users to search products using keywords such as medicine name, brand, or category.
- Filtering options must include:
 - Category
 - Manufacturer
 - Price range
 - Expiry period
 - Availability

6. Sorting by Price, Expiry Date, and Availability

- Users must be able to sort product listings based on:
 - Price (Low to High / High to Low)
 - Expiry date (Nearest expiry first)
 - Stock availability

7. Secure Authentication Using JWT

- The system must implement JSON Web Token (JWT) based authentication for all user sessions.
- JWT will ensure:
 - Secure user validation
 - Stateless authentication
 - Protection against unauthorized access
 - Role-based access control for Admin, Supplier, and Buyer

8. Dashboard for Users and Admin

- Supplier Dashboard:
 - Product management
 - Inventory overview
 - Orders received
 - Sales analytics
- Buyer Dashboard:
 - Orders placed
 - Order tracking
 - Purchase history
 - Invoice downloads
- Admin Dashboard:
 - User management
 - System monitoring
 - Transaction overview
 - Reports and analytics

9. Payment Processing System

- Integration with eSewa and Khalti APIs for real-time digital wallet payments
- Bank transfer option with manual or automated payment verification
- Payment status tracking (Pending, Completed, Failed) for every transaction
- Automated invoice generation after successful payment
- Payment history and financial reports accessible to users and admin
- Refund management system to handle failed or cancelled transactions with proper audit logs
- All financial transactions must be encrypted and comply with standard security practices

4.3.2. Non-Functional Requirements

Non-functional requirements define the quality attributes and constraints of the PharmaLink system, ensuring reliability, security, performance, and usability.

[i.]Performance Requirements

1.
 - The system shall support at least 1000+ concurrent users without performance degradation
 - Page load time shall not exceed 3 seconds under normal network conditions
 - Order processing and payment confirmation must occur in real time or near real time
 - The system shall be scalable to support increased users and transactions in the future

2. Availability and Reliability Requirements

- The system shall maintain an uptime of at least 99.5%
- Automatic system recovery shall be implemented in case of server failure
- Regular data backups shall be performed daily
- No loss of transaction data shall occur in the event of unexpected shutdowns

3. Security Requirements

- All user authentication must be implemented using JWT-based secure authentication
- All payment and personal data must be encrypted
- The system shall implement protection against common vulnerabilities (SQL injection, XSS, CSRF)
- Role-based access control must be enforced
- Compliance with OWASP security standards must be ensured

4. Usability Requirements

- The user interface shall be intuitive and easy to navigate
- Minimal training shall be required for new users
- The system shall provide meaningful error messages and user guidance
- Support for both desktop and mobile browsers (responsive design)

5. Compatibility Requirements

- Compatible with modern web browsers (Chrome, Firefox, Edge)
- Mobile-friendly and responsive UI
- Compatible with standard internet speeds available in Nepal

6. Maintainability Requirements

- Code shall follow modular architecture
- Proper documentation must be maintained
- The system shall support easy bug fixing and feature upgrades
- Logs shall be maintained for monitoring and troubleshooting

7. Legal and Regulatory Requirements

- Must adhere to Department of Drug Administration (DDA) Nepal regulations
- Must ensure secure handling of licensed pharmaceutical businesses only
- Data privacy must be maintained as per national cyber laws

4.4. Feasibility Study

4.4.1. Technical

PharmaLink will be developed using the **PERN stack**: PostgreSQL, Express.js, React.js, and Node.js. Raw SQL queries will be used for performance, with triggers and constraints ensuring data integrity. Secure payment gateways (eSewa, Khalti) will be integrated.

4.4.2. Operational

The system will be simple and user-friendly, ensuring smooth adoption by pharmacies and distributors with minimal technical knowledge.

4.4.3. Economic

Table 4.1: Sample Cost-Benefit Analysis of the Proposed Project

Item	Description	Cost (\$)	Benefit (\$)
Development Costs	Software Development	10,500	-
Hardware Costs	Servers and Equipment	3,500	-
Software Tools	Domain, SSL, APIs, Dev Tools	2,000	-
Maintenance Costs	Annual Maintenance	1,000	-
Total Costs		17,000	-
Increased Efficiency	Time Savings	-	20,000
Improved User Satisfaction	User Feedback	-	10,000
Revenue Increase	New Customers	-	20,000
Total Benefits		-	50,000
Net Benefit		17,000	33,000

4.4.4. Schedule

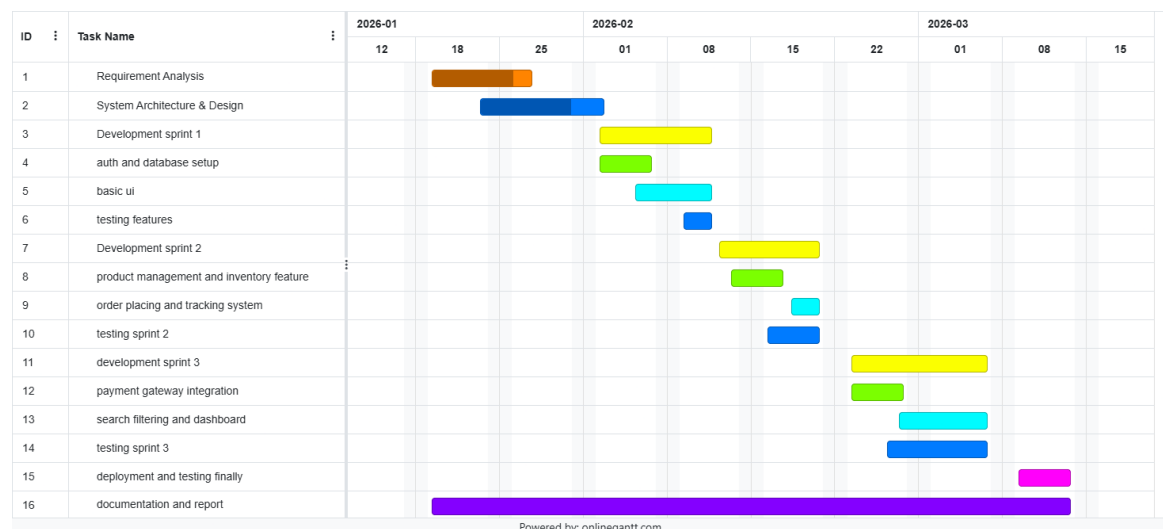


Figure 4.1: Gantt Chart demonstrating schedule feasibility

4.5. High-Level Design of System

PharmaLink's design is illustrated through two key diagrams:

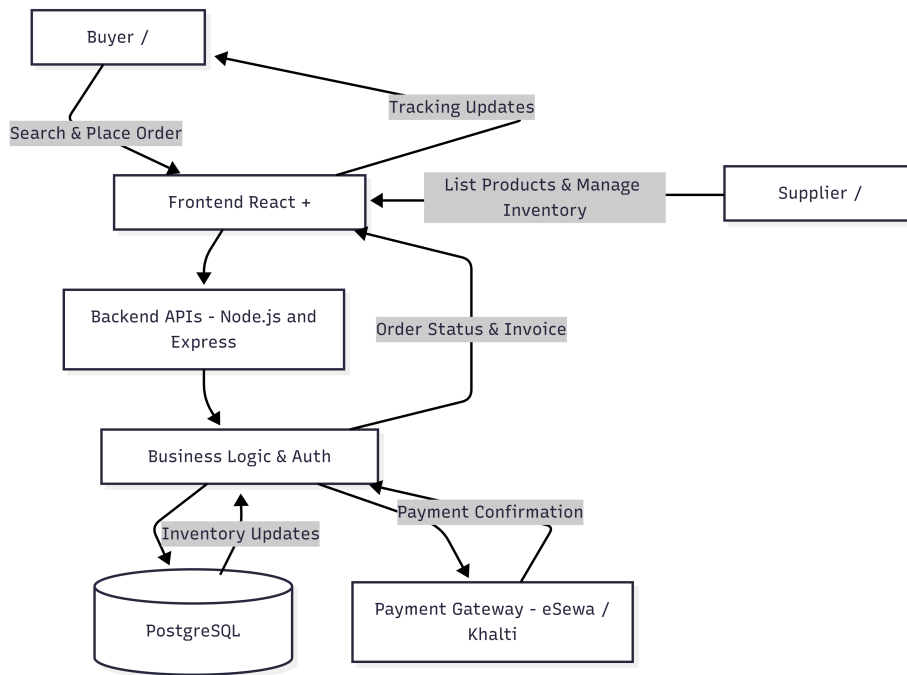


Figure 4.2: System Architecture Flow Diagram of PharmaLink

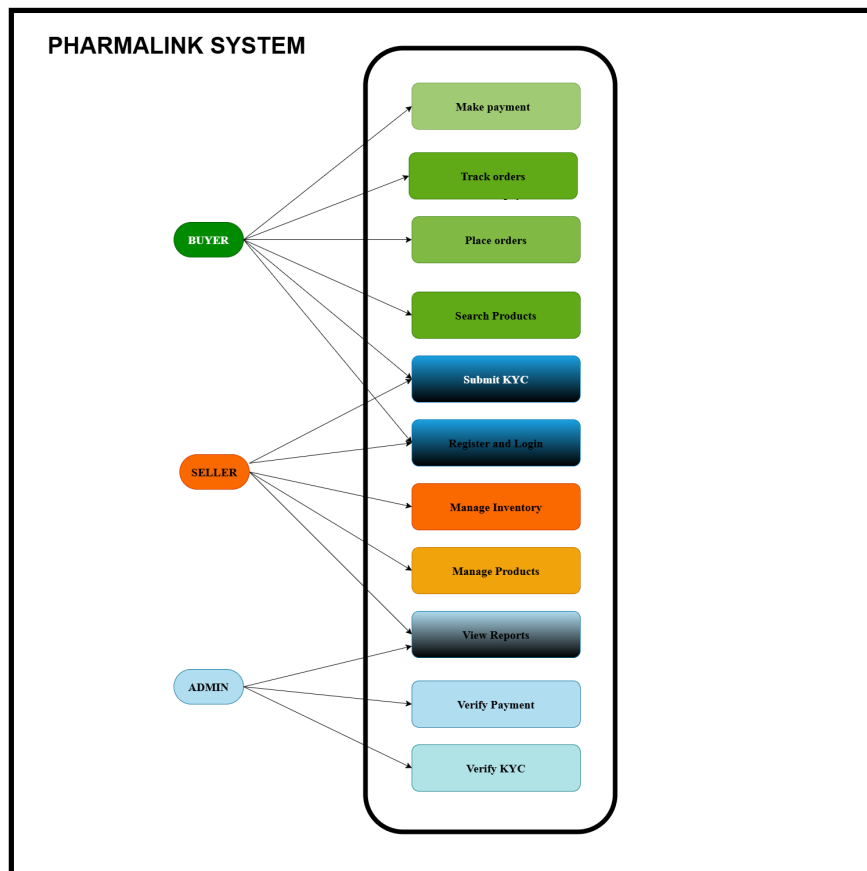


Figure 4.3: Use Case Diagram of PharmaLink

4.5.1. Architectural Approach

PharmaLink is based on a **Three-Tier Architecture**:

1. **Presentation Layer (Frontend)**: React.js interface for Admins, Suppliers, Buyers.
2. **Application Layer (Backend)**: Node.js + Express.js handling business logic, authentication, and APIs.
3. **Data Layer (Database)**: PostgreSQL storing users, products, orders, payments, and audit logs.

4.5.2. Design Principles

- Modularity
- Scalability
- Security by Design
- High Performance
- Maintainability

4.6. Description of Algorithms

This section outlines the core algorithms implemented in PharmaLink to ensure efficient data management, smart product search, automated inventory control, and secure payment processing.

4.6.1. CRUD-Based Product and Order Management

Algorithm Name: Create-Read-Update-Delete (CRUD)

Type: Traditional Data Management

Technology: Node.js + Express + PostgreSQL (Raw SQL)

Purpose: Efficient management of products, users, orders, and inventory.

Pseudocode (Simplified):


```

PROCEDURE CRUD_OPERATIONS(entity, operation, data):
    IF CREATE: VALIDATE data → INSERT
    IF READ:    SELECT records (with/without filter)
    IF UPDATE: IF record exists → UPDATE ELSE error
    IF DELETE: IF record exists & no dependencies → DELETE ELSE error
END PROCEDURE

```

4.6.2. Composition-Based Smart Search and Sorting

Algorithm Name: Inverted Index Search + Merge Sort

Type: Search and Sorting

Purpose: Enables buyers to find medicines by chemical composition and compare brands by price, expiry, or stock.

Concept: Build an inverted index mapping each composition (e.g., “Paracetamol 500mg”) to all products. On search, retrieve matches, apply filters, and sort results using Merge Sort.

Pseudocode (Simplified):

```

BUILD_INDEX():
    FOR product IN database:
        FOR comp IN product.compositions:
            key = NORMALIZE(comp)
            ADD product TO index[key]

SEARCH(query, sort_by, filters):
    result = index[query]
    APPLY filters (in_stock, max_price)
    SORT result BY sort_by USING MERGE_SORT
    RETURN result

```

4.6.3. Database-Level Automation

Algorithm Name: Event-Driven Trigger

Type: SQL Functions and Triggers

Purpose: Automates inventory updates, prevents invalid orders, and maintains consistency on cancellations.

Key Triggers:

- **Stock Reduction (After Insert):** Reduce stock on new order; notify supplier if below threshold.
- **Validation (Before Insert):** Block order if insufficient stock.
- **Restore Stock (After Update → Cancelled):** Restore stock when orders are cancelled; notify supplier.

Pseudocode (Simplified):

AFTER INSERT ON orders:

IF stock < qty → ROLLBACK ELSE UPDATE stock

BEFORE INSERT ON orders:

IF stock < qty → RAISE EXCEPTION

AFTER UPDATE ON orders:

IF status == CANCELLED → RESTORE stock

4.6.4. Payment Processing

Algorithm Name: HMAC-Based Secure Payment Verification

Type: Payment Gateway Integration + Cryptographic Verification

Purpose: Ensures secure transactions and authenticates payment callbacks from eSewa and Khalti.

Key Steps:

1. Generate transaction ID and store as pending.
2. Redirect to wallet gateway or bank transfer.
3. Verify callback using HMAC-SHA256 signature.
4. On success: update payment/order, generate invoice, notify users.
5. On failure: mark payment failed, notify buyer.
6. Bank transfer: admin verifies proof → approve or reject.

7. Auto-generate invoice when payment marked as PAID.

Pseudocode (Simplified):

```
PROCESS_PAYMENT(order, method, amount):  
    tx_id = UUID()  
    IF WALLET → DIGITAL_WALLET(tx_id)  
    ELSE BANK_TRANSFER(order)  
  
VERIFY_CALLBACK(data):  
    expected = HMAC_SHA256(data)  
    IF signature invalid → error  
    IF status == SUCCESS → ON_SUCCESS(tx_id)  
    ELSE ON_FAILURE(tx_id)
```

4.6.5. Summary of Algorithms

Algorithm	Name	Time Complexity
A. CRUD	Create-Read-Update-Delete	$O(1)$ per operation (with indexing)
B. Search & Sort	Inverted Index + Merge Sort	$O(n \log n)$
C. Automation	Event-Driven Trigger	$O(k)$ per trigger (k = order items)
D. Payment	HMAC-SHA256 Verification	$O(1)$ signature verification

Table 4.2: Summary of Algorithms Used in PharmaLink

5. Expected Output

The proposed PharmaLink system is expected to deliver the following outcomes:

1. A fully functional B2B pharmaceutical e-commerce platform.
2. Digital product listing and streamlined order processing.
3. Improved inventory control with real-time visibility.
4. Faster and more transparent pharmaceutical trade.
5. A secure and scalable system architecture.
6. Reduced manual errors through automation.
7. Secure and automated payment processing.
8. Multiple payment options suitable for the Nepali market (eSewa, Khalti, bank transfers).
9. Reduced payment-related disputes and delays.
10. Automated financial record keeping and reporting.

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